THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 120 cm	1.2 x 100 = 120.
2	**(a)** cork and leaf	Cork and leaf are lighter than water.
3	**(c)** fever	Above 37 °C.
4	**(b)** evaporation	Slow vaporisation.
5	**(d)** a mass vibrating on a spring	It moves to and fro on a straight path.
6	**(c)** the boat's shape holds a lot of air	Large volume with air makes the boat's overall density small.
7	**(c)** because of the kink	Kink holds mercury.
8	**(c)** evaporation and condensation	Water vapour cools.
9	**(d)** hold the tape against the edge of the object so that measuring starts from zero	Hook sets zero.
10	**(d)** food does not stick to the surface	Non-stick.
11	**(b)** 30 °C	Room temperature.
12	**(b)** drip irrigation	Saves water.
13	**(b)** 3.5 km	Divide by 1000.
14	**(b)** salt and sugar	Both dissolve.
15	**(d)** below 35 °C	So the reading is accurate.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	3.75 x 1000 = 3750 m (1); 3750 x 100 = 375000 cm (1).	1+1
17	The boat has a large hollow shape with air (1); Water supports it as its average density is low (1).	1+1
18	Above normal 37 °C (1); The person has a fever (1).	1+1
19	Heat is used to change water into steam (1); It is not used to raise the temperature (1).	1+1
20	Rotation: spinning about own axis, like Earth in a day (1); Revolution: going around another body, like Earth around the Sun (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	240 cm / 15 = 16 (1); 16 pieces exactly (1); Leftover 0 cm (1).	1+1+1
22	Magnet removes iron (1); Dissolve sugar in water (1); Filter sand and evaporate water for sugar (1).	1+1+1
23	Heat flows to room (1); Water cools (1); Becomes about 25 °C (1).	1+1+1
24	Evaporation from the sea (1); Condensation into clouds (1); Precipitation and flow back (1).	1+1+1
	OR: T = 42/21 = 2 s (1); 60 s / 2 s (1); 30 oscillations (1).	1+1+1
25	Rough mat cleans shoes (1); Smooth top is easy to clean (1); Window lets in light (1).	1+1+1
	OR: Shiny walls reflect heat back into the flask (1); Less heat escapes (1); Keeps drink hot or cold (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Wrap thread around the glass (1); Measure its length (1); Measure the diameter with a scale (1); Divide circumference by diameter (1); Ratio is nearly 3.14 (1).	1+1+1+1+1
	OR: Heat water and cover (1); Measure temperature drop after 30 min (1); Same conditions for all (1); The least drop is best insulator (1); Cork or plastic likely (1).	1+1+1+1+1
27	Rise = 8 °C each (1); After 2 min: 38 (1); After 4 min: 46 (1); After 10 min = 30 + 5 x 8 (1); 70 °C (1).	1+1+1+1+1

	OR: Take plates of different size with same water (1); Keep one in sun and one in shade (1); Use a fan for one (1); Note time to dry (1); Large, warm, windy dries fastest (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** boiling	Vaporisation.
2	**(b)** circular motion	Path is circular.
3	**(a)** wax melts quickly along the rod when one end is heated	Heat travels along it.
4	**(b)** bulb is sensitive to body heat	Hand heat changes the reading.
5	**(a)** ice – solid	Ice is solid.
6	**(a)** easier to read correctly	Accuracy.
7	**(a)** wool is a poor conductor of heat	Traps body heat.
8	**(b)** 40 °C	Between the two.
9	**(a)** rainwater harvesting	Water seeps into soil.
10	**(c)** 50 cm	250 / 5 = 50.
11	**(d)** metal	Metals have these properties.
12	**(a)** iron	Iron conducts heat quickly to hand.
13	**(d)** water boils	Boiling point.
14	**(a)** rotational motion	Rotates about its axis.
15	**(a)** a thick wooden handle	Wood does not conduct heat.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	No privacy (1); Frosted translucent glass is better (1).	1+1
17	Air trapped between layers (1); Air insulates (1).	1+1
18	More heat flows in (1); Ice absorbs heat and melts (1).	1+1
19	Time period = total time / oscillations (1); 60/30 = 2 s (1).	1+1
20	Both disappear on stirring (1); Both are soluble (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Only water evaporates (1); Salts remain (1); Condensed water is fresh (1).	1+1+1
22	15 x 1000 = 15000 m (1); 15000 x 100 (1); 1500000 cm (1).	1+1+1
23	Thermocol is light (1); Combined object has low overall density (1); It floats (1).	1+1+1
24	Allows heat to escape (1); Absorbs sweat (1); Keeps cool (1).	1+1+1
	OR: Irrigation (1); Electricity (1); Flood control (1).	1+1+1
25	1.1 m = 110 cm (1); 110 / 5 (1); 22 cm (1).	1+1+1
	OR: Heat one end of each (1); Place wax pins at equal distance (1); The pin that falls first shows the best conductor (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Passenger at rest wrt train (1); Trees seem to move backward (1); Sun seems to move across the sky (1); Earth's rotation causes it (1); Depends on frame (1).	1+1+1+1+1
	OR: Dissolve salt in water (1); Fold filter paper (1); Pour mixture (1); Sand stays on paper (1); Salt solution passes (1).	1+1+1+1+1

27	Hot water at 70 °C, cold water at 20 °C (1); Heat flows from hot to cold (1); Final in between (1); No more flow once equal (1); Same temperature (1).	1+1+1+1+1
	OR: Filter through cloth (1); Boil for 10 minutes (1); Cool (1); Store in a clean covered vessel (1); Kills germs (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** water evaporating from the skin takes heat away	Evaporation cools.
2	**(d)** heat is used to change the state	Energy is used for melting.
3	**(b)** 0.24 m	Divide by 100.
4	**(b)** steel	Strong and waterproof.
5	**(c)** the metal conducts heat to the hand faster	Rate of conduction.
6	**(d)** roots hold soil and water	Roots help soak water.
7	**(d)** 20 cm	80 / 4 = 20.
8	**(a)** thin cotton cloth	Translucent cloth.
9	**(b)** surroundings	Heat flows from higher temperature.
10	**(d)** slows down	Humid air.
11	**(c)** orbital (revolution)	Revolution.
12	**(a)** wooden chips	Wood is non-magnetic.
13	**(a)** it can break	High temperature breaks it.
14	**(a)** heat is used for changing water to steam	Heat used in vaporisation.
15	**(a)** 5	150/30.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Keeping tea hot (1); Keeping cold drinks cool (1).	1+1
17	Evaporation through pores cools it (1); Heat taken away (1).	1+1
18	4.5 m = 450 cm (1); 450 / 25 = 18 pieces (1).	1+1
19	Steel is stronger and harder than pure iron (1); Stainless steel does not rust easily (1).	1+1
20	Bottom is hotter than the liquid (1); It would give a wrong reading (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Heat one end of each (1); Place wax pins (1); Compare which falls first (1).	1+1+1
22	Put a few drops of spirit on the hand (1); It evaporates quickly (1); Hand feels cool (1).	1+1+1
23	Wrap round the waist (1); Keep tape flat (1); Read where ends meet (1).	1+1+1
24	Cut a piece of cork (1); Even small solid pieces float (1); Hence lighter material (1).	1+1+1
	OR: Iron conducts heat quickly (1); Wood conducts poorly (1); Rate differs (1).	1+1+1
25	Sunny: fast (1); Cloudy: slow (1); Heat increases evaporation (1).	1+1+1
	OR: Pen: cm (1); River: km (1); Coin: mm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Sum = 601.5 (1); Mean = 601.5/5 = 120.3 cm (1); Repeating reduces errors (1); Readings vary slightly (1); Average is more reliable (1).	1+1+1+1+1
	OR: Heat a rod for conduction (1); Stir in water for solubility (1); Torch for transparency (1); Put in water for floating (1); Scratch for hardness (1).	1+1+1+1+1
27	Mercury expands (1); Rises in the narrow tube (1); Kink prevents falling (1); Heating would break it (1); Range is limited (1).	1+1+1+1+1

	OR: Sun heats the sea (1); Vapour rises (1); Condenses into clouds (1); Droplets grow (1); Fall as rain (1).	1+1+1+1+1

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